

Ramkrishna Acharya

Machine Learning Engineer | Computer Vision | Applied AI
Erlangen, Bavaria – Germany

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Nationality: Nepalese

Industry Profile

ML engineer with 4+ years of experience in industrial computer vision, anomaly detection, OCR, security-focused ML, synthetic data, and data-platform engineering. Proven track record building PoCs, evaluation pipelines, and reusable ML tools with Python, PyTorch, OpenCV, FastAPI, SQL, Docker, ONNX/TensorRT, and MLOps workflows.

Core Strengths

Applied ML / CV Anomaly detection, OCR, segmentation, object detection, image processing, video analysis, synthetic data
Reliable AI Adversarial ML, explainable AI, intrusion detection, novel attack detection, robust cyber-physical systems
ML Engineering FastAPI, model evaluation pipelines, SQL-backed tooling, Dockerized workflows, ONNX/TensorRT deployment

Recent Experiences

Upwork

Freelance Data Scientist (Profile)

Remote
May 2026 – Present

- Developing real-time computer vision solutions for anomaly detection and deployable surveillance systems
- Stack:** Python, OpenCV, Django, PyTorch, Docker, Git.

Primetals Technologies Germany GmbH

Working Student – Computer Vision

Erlangen, Germany
May 2023 – Mar. 2026

- Built proof-of-concept anomaly detection frameworks using deep learning techniques
- Benchmarked and trained state-of-the-art vision-language models for OCR in industrial environments
- Built model evaluation framework using FastAPI and SQL, saving 100+ hours of video analysis time
- Designed a keypoint-based algorithm for camera vibration detection, improving prediction precision under noisy conditions
- Developed synthetic data generators for OCR and segmentation, improving model training under limited data
- Stack:** PyTorch, MMEEngine, OpenCV, TensorRT, ONNX, MLflow, MQTT

Chair for Computer Science 7, FAU Erlangen-Nürnberg

Student Assistant – Cybersecurity Fundamentals in Networking

Erlangen, Germany
Oct. 2025 – Mar. 2026

- Prepared exercise material and tutored students in the course *Cybersecurity Fundamentals in Networking*
- Supported hands-on labs using VirtualBox, Kali Linux, Wireshark, and ScaPy.

Older Experiences

- Upwork**, Freelance Data Science Engineer, Remote Jan. 2022 – Oct. 2022
Built real-time alert-based strategy backtesting platforms to reduce options and stock trading risk using Python, AsyncIO, WebSockets
- Extensodata Pvt. Ltd.**, Associate Data Scientist, Kathmandu, Nepal Aug. 2021 – Aug. 2022
Built data pipelines and intelligent algorithms for fintech systems, generating insights to support loan-default reduction using Python, MySQL, Pentaho, Apache Airflow, Apache Superset, and Git.
- diyo.ai**, Unity3D Development Intern, Kathmandu, Nepal Nov. 2020 – Mar. 2021
Built an AR jewelry visualization application using Unity3D, Google ARCore, C#, and Git.
- MPercept Technology**, AI Developer Intern, Kathmandu, Nepal Aug. 2019 – Jan. 2020
Developed Rasa-based conversational agents for e-commerce and travel platforms using Python, OpenCV, AWS EC2, and Git.

Education

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)

M.Sc. Data Science, German grade scale: 1.8 (Good)

Erlangen, Germany
2022 – 2025

Thesis: *Enhancing Smart Grid Security: A Deep Learning Approach to Adversarial Intrusion Detection.*

Focus: Applied machine learning, deep learning, adversarial robustness.

Tribhuvan University

B.Sc. Computer Science and Information Technology, grade: 72.07/100 (First Division)

Kathmandu, Nepal
2015 – 2019

Research Project: *Devanagari Handwritten Character Recognition.*

Focus: Image processing and neural networks.

Publications

- Ramkrishna Acharya**, Loui Al Sardy, Mamdouh Muhammad, Reinhard German. *ADVIS-G: An Adversarially Defended Intrusion Detection System for Smart Grids Using Deep Learning*. KI – Künstliche Intelligenz, 2026. DOI | Code
- Ramkrishna Acharya**, Loui Al Sardy, Mamdouh Muhammad. *ROSAID-ST: Adversarially Robust Intrusion and Anomaly Detection in ROS Networks via Student-Teacher Deep Learning*. under review, 2026. Journal | Code

Selected Projects

- Session Feature Extractor** (2026): PyPI tool for extracting image-style features from PCAP traffic for intrusion-detection.
- Image Baker** (2025): PyPI tool for annotation, model-assisted labeling, and synthetic image data generation.
- SmokeSim** (2024): PyPI package for smoke simulation, image augmentation, and segmentation-data generation.
- Neural Template Matching** (2024): An experiment for DL based template matching on the COCO dataset using HPC.

Technical Skills

Programming Python advanced, SQL, Bash/Shell, JavaScript basic, C# basic
ML / CV PyTorch, TensorFlow, Keras, Scikit-learn, Kornia, HuggingFace, MMEEngine, OpenCV, RoboFlow
MLOps / Tools Docker, Git/GitHub, MLflow, DVC, ONNX, TensorRT, Apache Airflow, Pentaho, MQTT
Systems / Data FastAPI, PySide6, WebSockets, PostgreSQL, Firestore, AWS EC2, GitHub Actions, Apache Superset
Languages English (C1), German (A2), Nepali (Native)